Routing Security

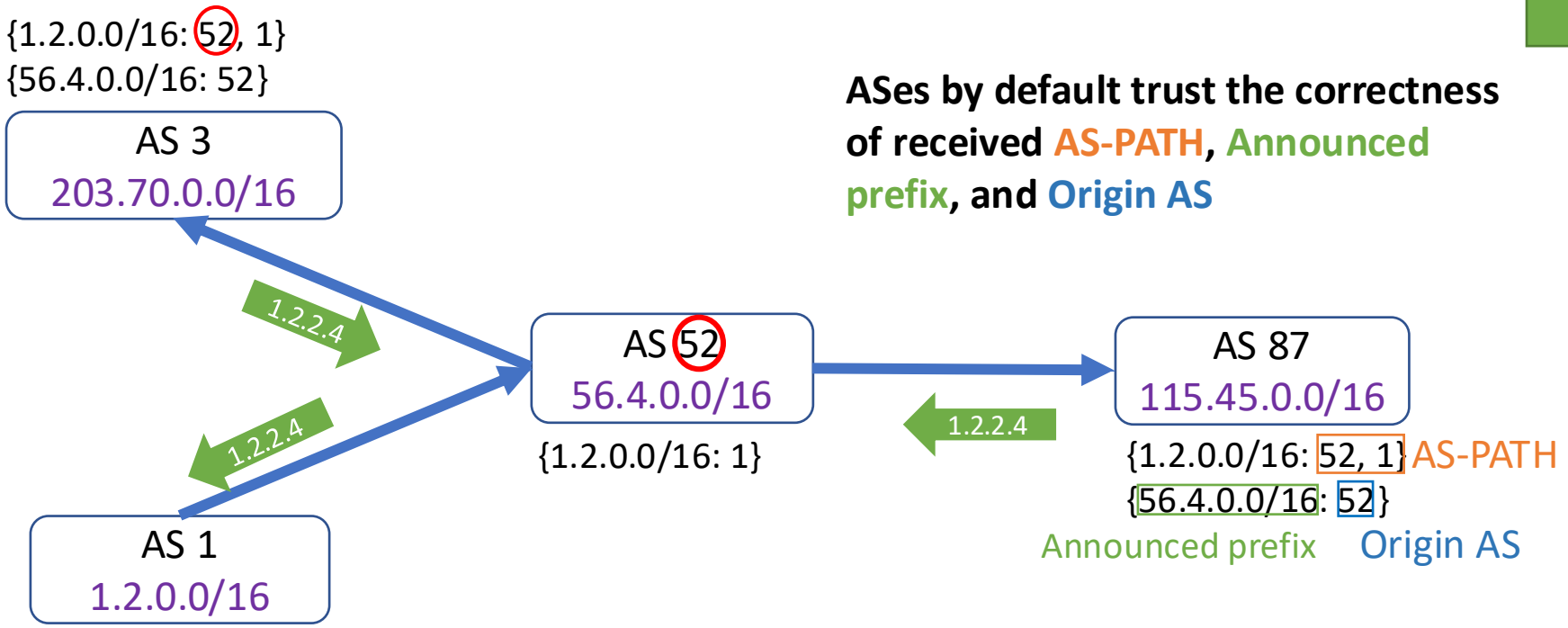
4/8/2026

ASes Share Reachability Information via BGP

- ASes send BGP messages to each other (control plane)
- Traffic flow in the opposite direction (data plane)

Where do I send an IP packet destined to 1.2.2.4?

ASes by default trust the correctness of received AS-PATH, Announced prefix, and Origin AS



BGP Vulnerability

- BGP includes no mechanism to validate the correctness of the information exchanged between participants.
  - Attackers can originate any prefix without authorization (origin hijack)
  - Attackers can modify any part of received BGP messages (path hijack)
  - Misconfigurations can cause damage (route leak)

- BGP hijacks - **malicious re-routing** of Internet **traffic** across network boundaries

Harms of BGP Hijacks

- Interrupted service
  - Victim may become unreachable
- Compromised privacy
  - Attackers eavesdrop on your activities over the Internet
- Stolen Identity
  - Attackers use victim’s identity to conduct illegal activities such as phishing and spamming
- Monetary loss
  - Cryptocurrency hijacks

BGP hijacking on the news

BGP Hijack of Amazon DNS to Steal Crypto Currency

Research // Apr 25, 2018 // Doug Madory

THANKS, BGP. —

BGP event sends European mobile traffic through China Telecom for 2 hours

Improper leak to Chinese-government-owned telecom lasts up to two hours.

DAN GOODIN - 6/8/2019, 9:05 AM

BORDER GATEWAY PROTOCOL —

How 3ve's BGP hijackers eluded the Internet—and made $29M

3ve used addresses of unsuspecting owners—like the US Air Force.

DAN GOODIN - 12/21/2018, 9:30 AM

Pakistan hijacks YouTube

Research // Feb 24, 2008 // Dyn Guest Blogs

Crypto Exchange KLAYswap Loses $1.9M After BGP Hijack

Hackers Performed Border Gateway Protocol Hack to Conduct Illegal Transactions

Prajeet Nair (@prajeetspeaks) • February 16, 2022

Cloudflare Confirms Recent 1.1.1.1 DNS Outage Caused by BGP Attack or Hijack

By Guru Baran - July 17, 2025

Types of Hijacks

{1.2.0.0/16: 52, 1} AS-PATH
{56.4.0.0/16: 52}
Announced prefix Origin AS

- Origin hijack
  - Exact prefix hijack
  - Sub-prefix hijack
  - Attacker changes the origin AS and/or the announced prefix

- AS-PATH manipulation
  - Attacker changes the one or more hops in AS-PATH

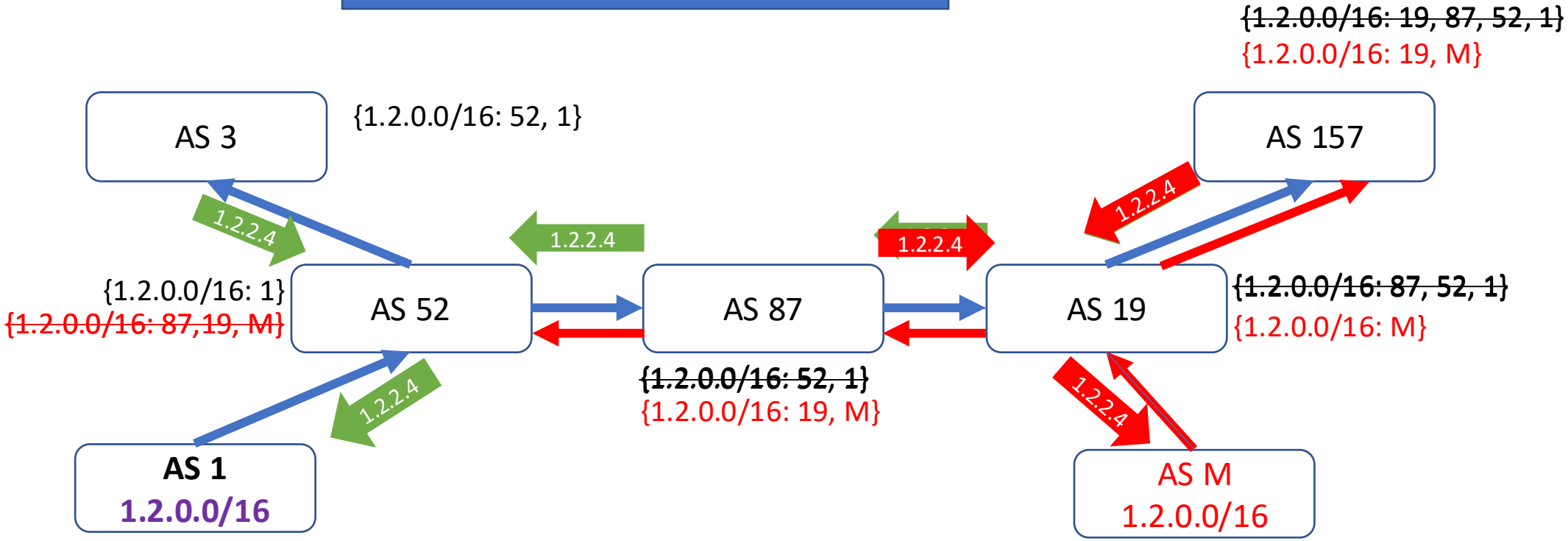
Man in the Middle!

Origin Hijack - attacker changes the origin AS of a prefix to its own

- Exact-prefix hijacking
  - Hijacker announces the victim’s prefix

Origin Hijack - Exact Prefix Hijacking

Route selection:
Shortest AS-PATH preferred!

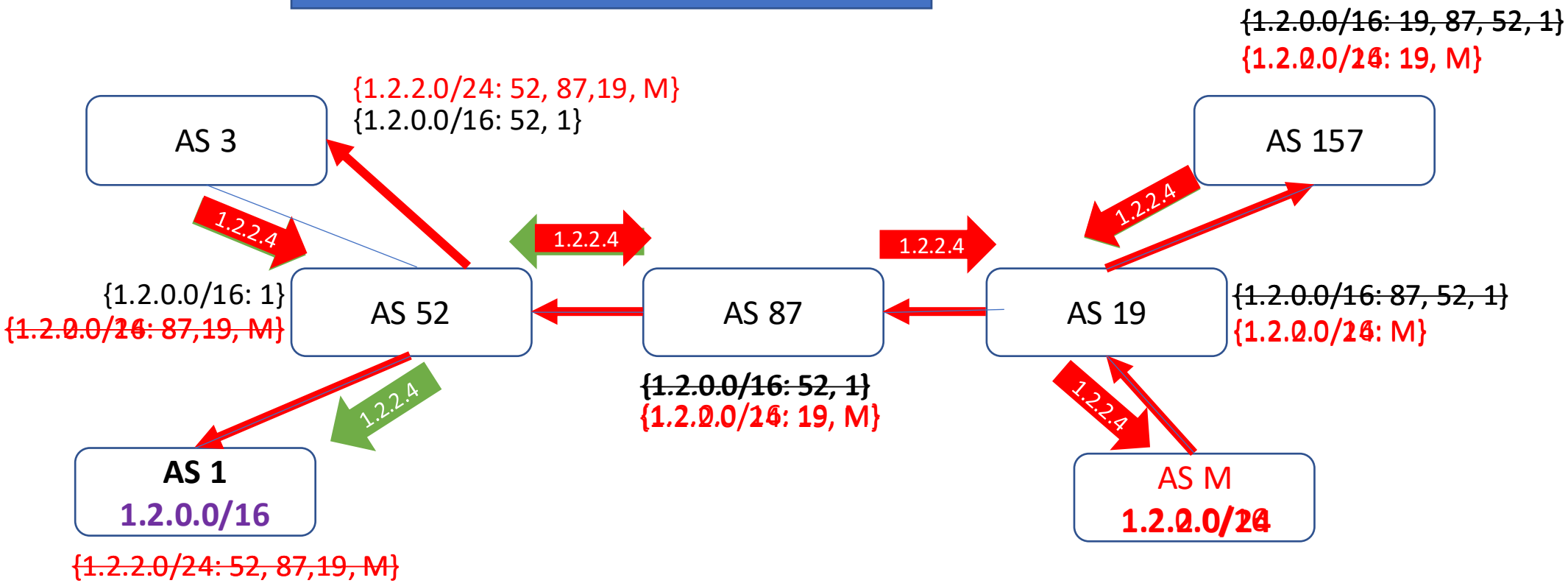


Origin Hijack – Exact Prefix Hijacking

- What malicious activity can it be used for?
- What are the limitations of exact-prefix hijacking?

Origin Hijack - Sub-prefix Hijacking

Route selection:
Longest prefix matching!

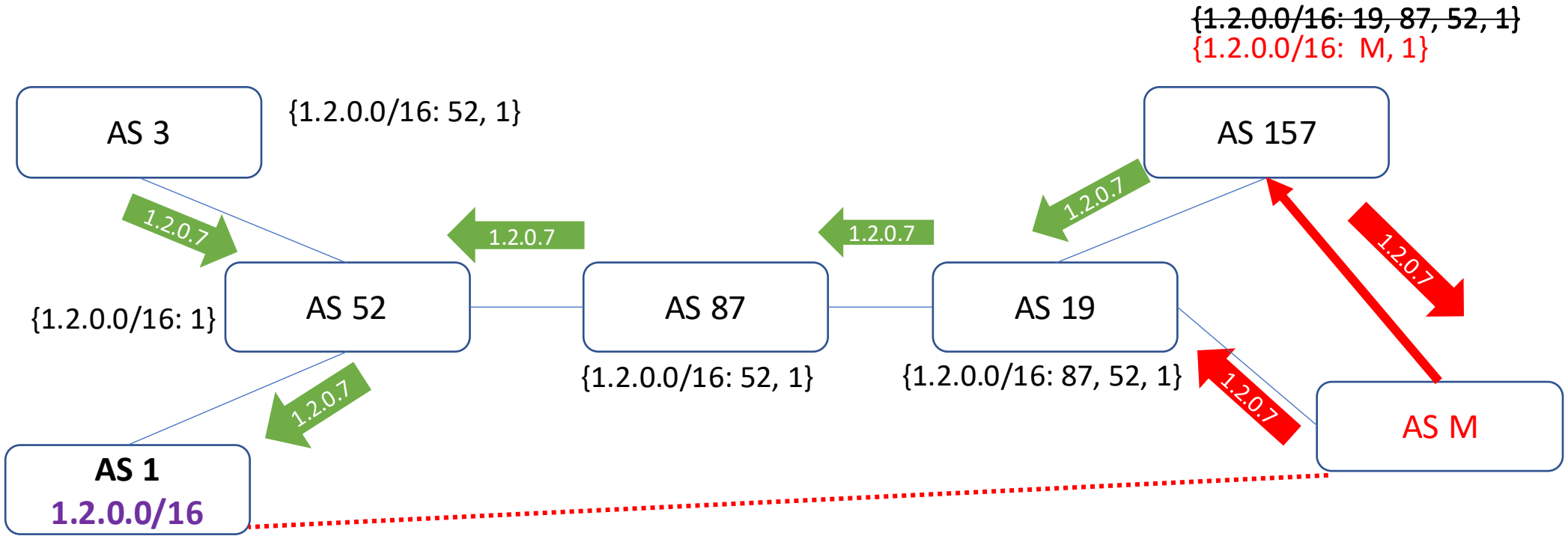


Origin Hijack – Sub-prefix Hijacking

- What can sub-prefix hijacking be used for?
- What are the limitations of sub-prefix hijacking?

AS-PATH Hijacking (Man in the Middle)

- Attacker changes the one or more hops in AS-PATH



Trust problem in BGP

- BGP messages received from neighbors cannot always be trusted
- Who can we trust?

Authoritative Databases

- Authoritative databases for verifying BGP origin information

- Internet Routing Registry (IRR)
  - First introduced in 1995
- Resource Public Key Infrastructure (RPKI)
  - First introduced in 2012

Internet Routing Registry

- Designed for sharing routing policy
- **25** IRR databases maintained by different organizations
  - Regional Internet Registries (RIRs), LIRs
  - Non-profit orgs
  - Commercial companies

Example objects in the IRR database

- *mntner*
  - The *mntner* class specifies authentication information required to create, delete and update RPSL objects.

```
mntner:      RIPE-NCC-MNT
descr:       RIPE-NCC Maintainer
admin-c:     DK58
tech-c:      OPS4-RIPE
upd-to:      ops@ripe.net
mnt-nfy:     ops-fyi@ripe.net
auth:        CRYPT-PW lz1A7/JnfkTtI
mnt-by:      RIPE-NCC-MNT
changed:     ripe-dbm@ripe.net 19970820
source:      RIPE
```

Example objects in the IRR database

- *route*
  - Each interAS route (also referred to as an interdomain route) originated by an AS is specified using a route object. – RFC2622
  - route and origin fields

```
route:          167.96.0.0/16
descr:          Cyprus Minerals Company
                9100 East Mineral Circle
                M.S. 257N
                Englewood
                CO 80112, USA
origin:         AS2900
member-of:      RS-COMM_NSFNET
mnt-by:         MAINT-AS2900
changed:        nsfnet-admin@merit.edu 19950506
source:         RADB
```

IRR records can be falsified…

- In 2020, gohosted.eu registered 4 UCSD-originated prefixes in RADB.
- gohosted.eu originated those prefixes in BGP until early 2021 (BGP hijacking) .
- The hijacker’s upstream providers accepted the BGP announcements because they matched RADB records.

```
route:      44.190.131.0/24
origin:     AS207427
descr:      NLVIX
mnt-by:     MAINT-213282
changed:    info@webdiensten.nu 20201216  #18:41:36Z
source:     RADB
```

RPKI

Resource Public Key Infrastructure (RPKI)

- The Resource Public Key Infrastructure (RPKI) is designed to improve routing security
- RPKI proves the authorization of ASes to use their prefixes
  - Essentially, RPKI is a cryptographically verifiable Prefix-to-AS database.
- Network operators prioritize RPKI over other sources of information to validate BGP information.

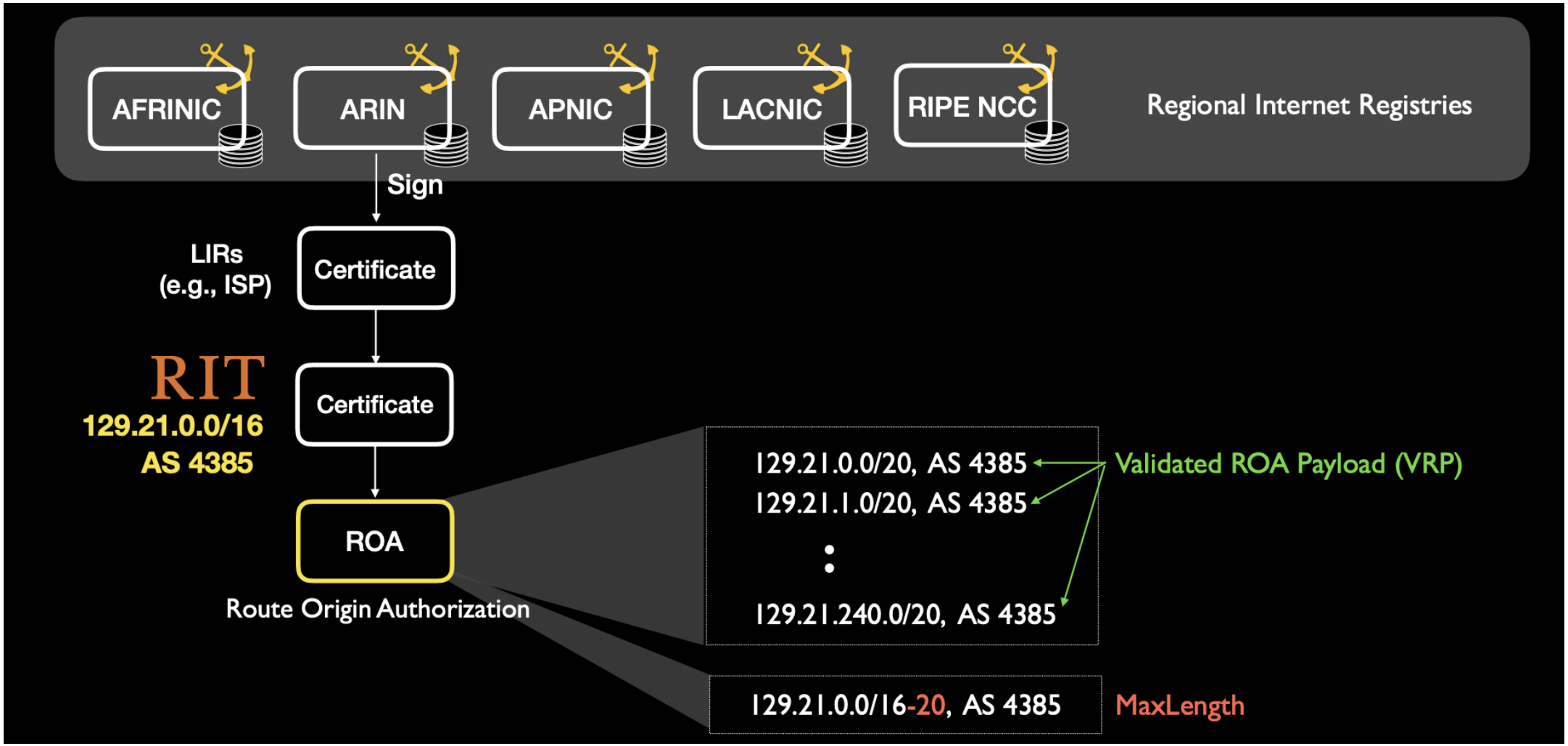
Purpose of RPKI

- Prevent BGP hijacking
  - Which types of BGP hijacking does is RPKI designed to prevent?
    - **Exact Prefix hijacking**
    - **Sub-prefix hijacking**
    - AS Path hijacking (nope)
- Prevent misconfiguration
  - Fat-finger errors

RPKI objects

- Certificate Authority certificate
  - Binds IP prefixes to AS numbers
- Route Origin Authorization (ROA)
  - Authorizes an AS to announce certain IP prefixes in BGP

RPKI Hierarchy



*Slide adapted from Chung et al. "RPKI is Coming of Age" IMC 2019.

Validated ROA Payload (VRP)

- A set of tuples (ASN, IP prefix, max length) constructed from parsing ROAs

```
,URI,ASN,IP Prefix,Max Length,Not Before,Not After
0,rsync://rpki.afrinic.net/repository/member_repository/F36FEA6B/7B27920E8B8211E9888B6F73F8AEA228/DB963EBAA95011E9B922BA68F8
AEA228.roa,AS0,2001:43f8::/48,128,2019-07-18 11:40:32,2029-07-18 11:40:32
1,rsync://rpki.afrinic.net/repository/member_repository/F36FEA6B/7B27920E8B8211E9888B6F73F8AEA228/7B3DF01C8F6F11E9B8F82D5FF8
AEA228.roa,AS327844,2001:43f8:1::/48,128,2019-06-15 13:14:13,2029-06-12 13:14:13
2,rsync://rpki.afrinic.net/repository/member_repository/F369B918/AE6F6A464B4911EB8D665384F8AEA228/5784F80C4B5011EB9EB3220CF8
AEA228.roa,AS37062,2001:43f8:400::/40,128,2020-12-31 10:09:57,2023-12-31 10:09:57
3,rsync://rpki.afrinic.net/repository/member_repository/F365C79E/1FC34BE6BEDA11EB8F5CE243F8AEA228/C0526AECBEDA11EB9591CE44F8
AEA228.roa,AS328732,2c0f:e9a8::/32,128,2021-05-27 11:00:26,2022-05-27 11:00:26
```

Max Length

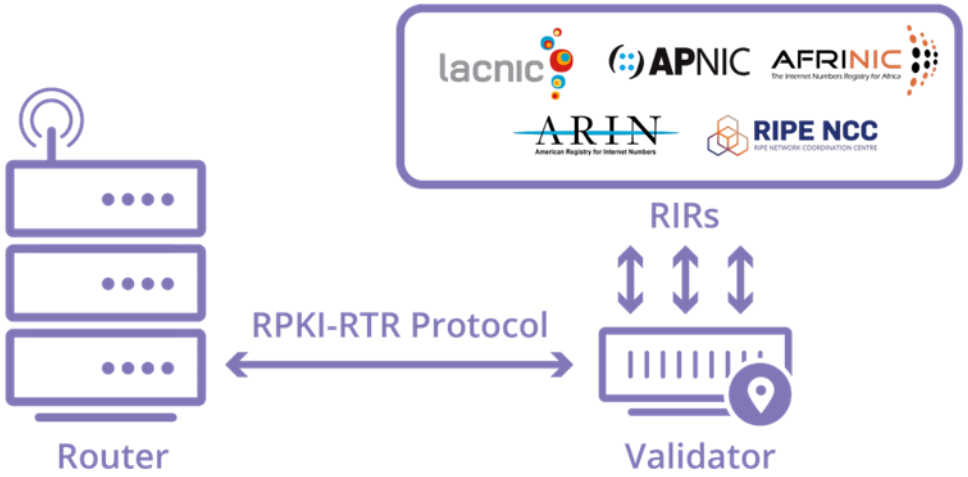
- Specifies the longest prefix length for the authorized IP prefix
- Used when an AS wants to de-aggregate a prefix

- Example: De-aggregating 129.21.0.0/16 into multiple /20 blocks

VRP: 129.21.0.0/16-20, AS 4385

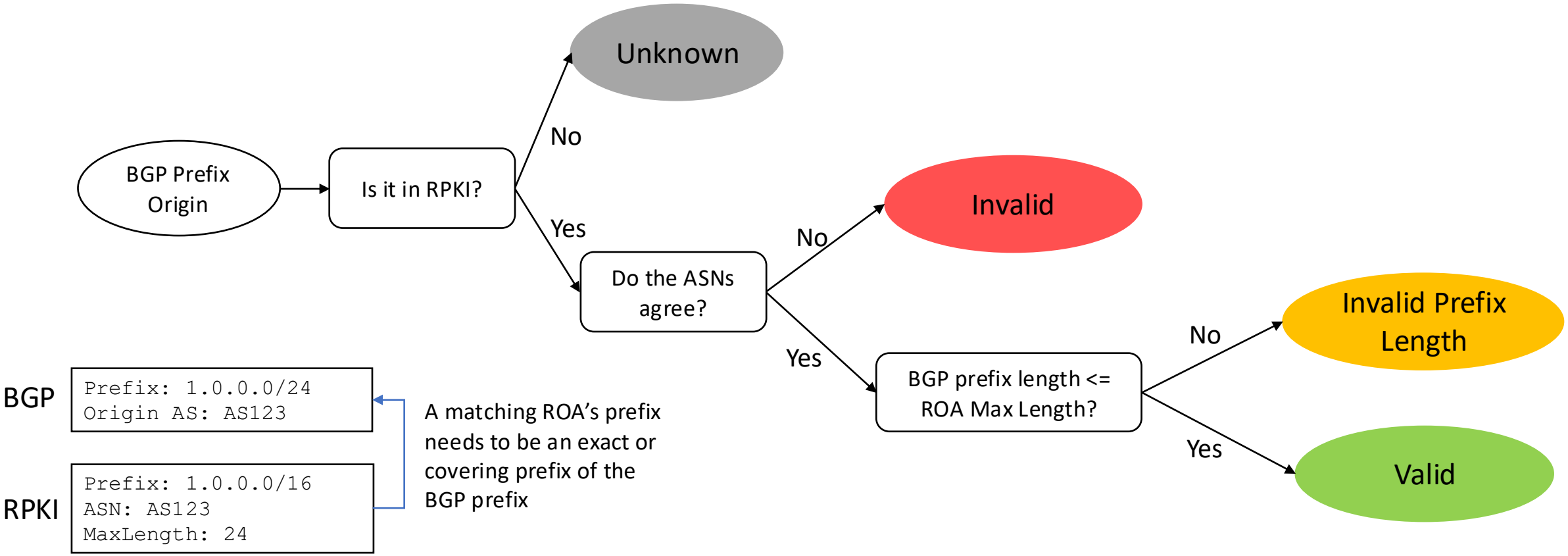
Route Origin Validation (ROV)

- Networks use Relying Party (RP) software to retrieve and validate RPKI objects from repositories
  - Construct VRPs
- Network then validate BGP origin information against VRPs

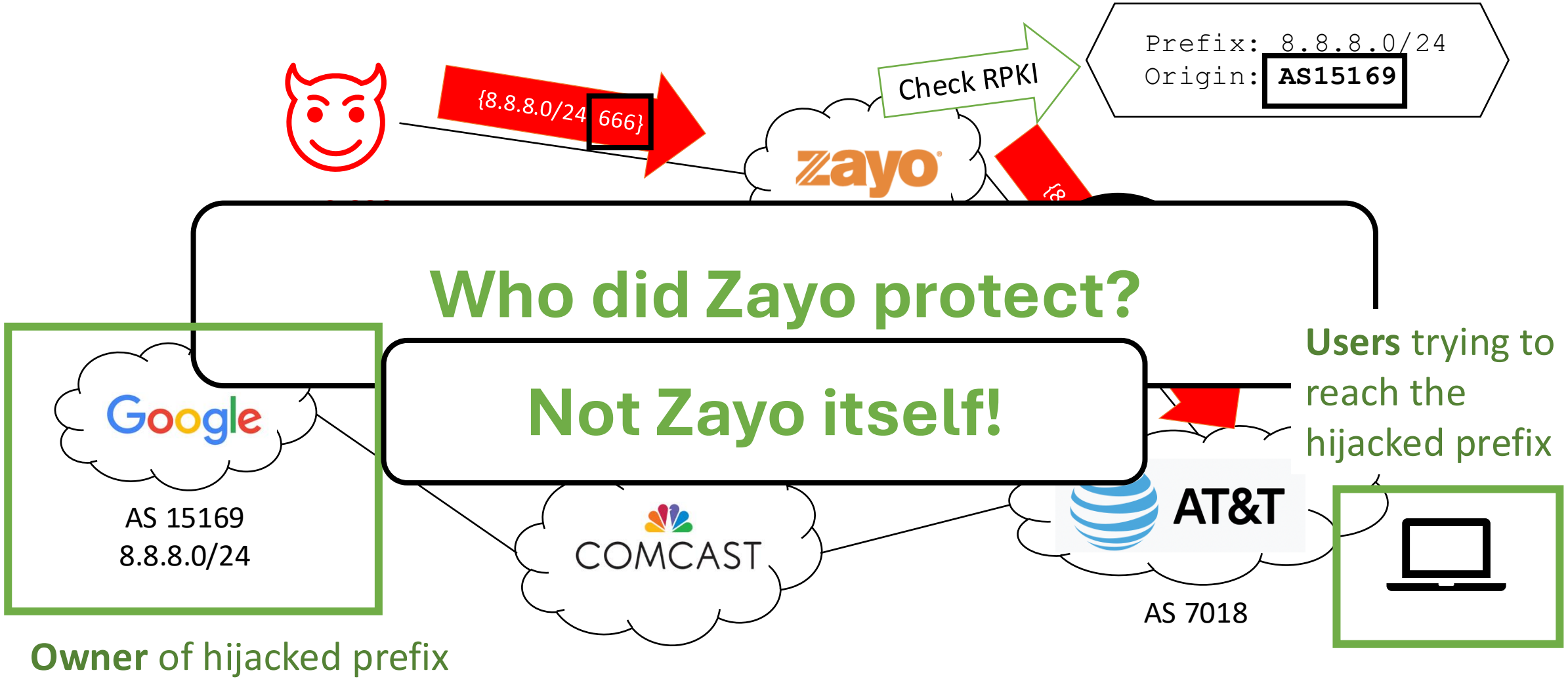


*Picture from Cloudflare

Validating BGP Announcements

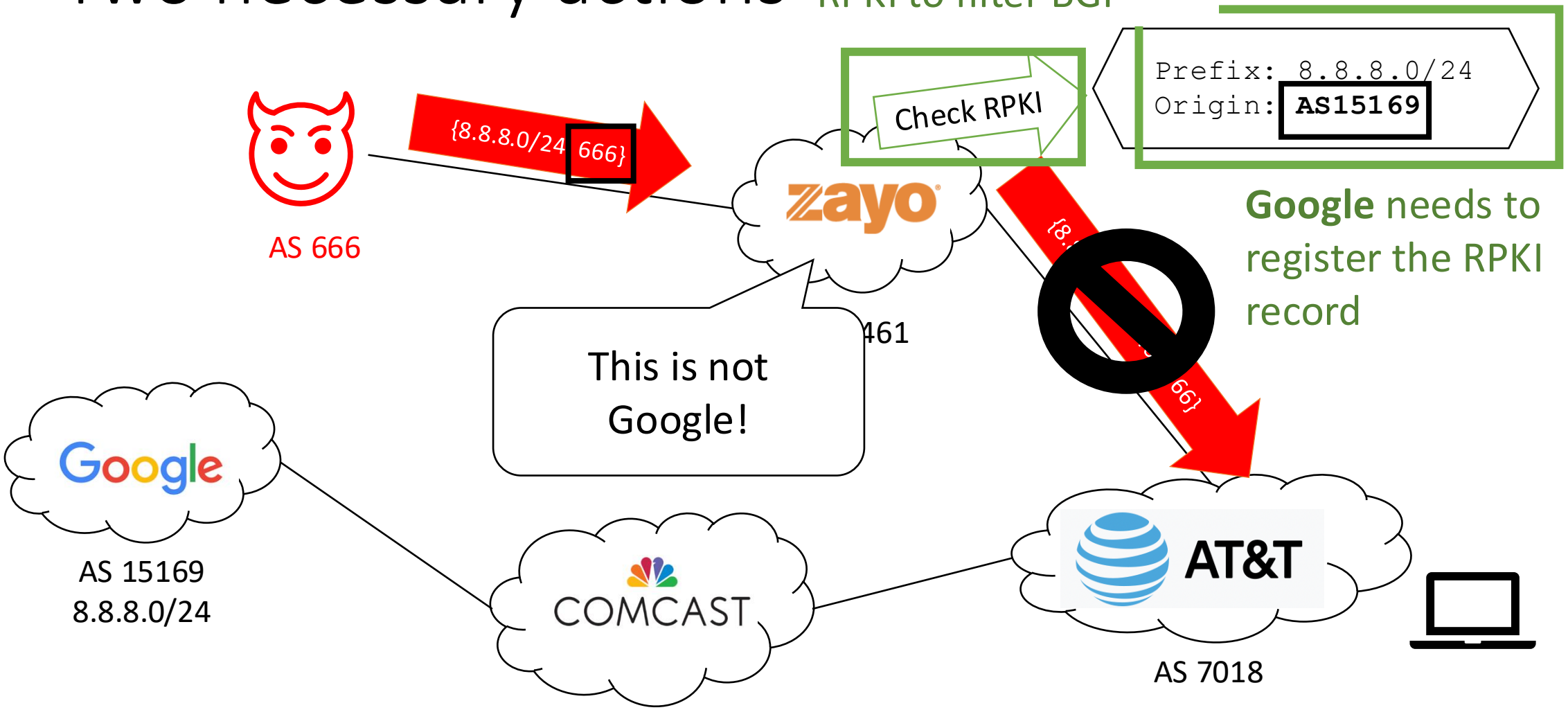


Incentive problem



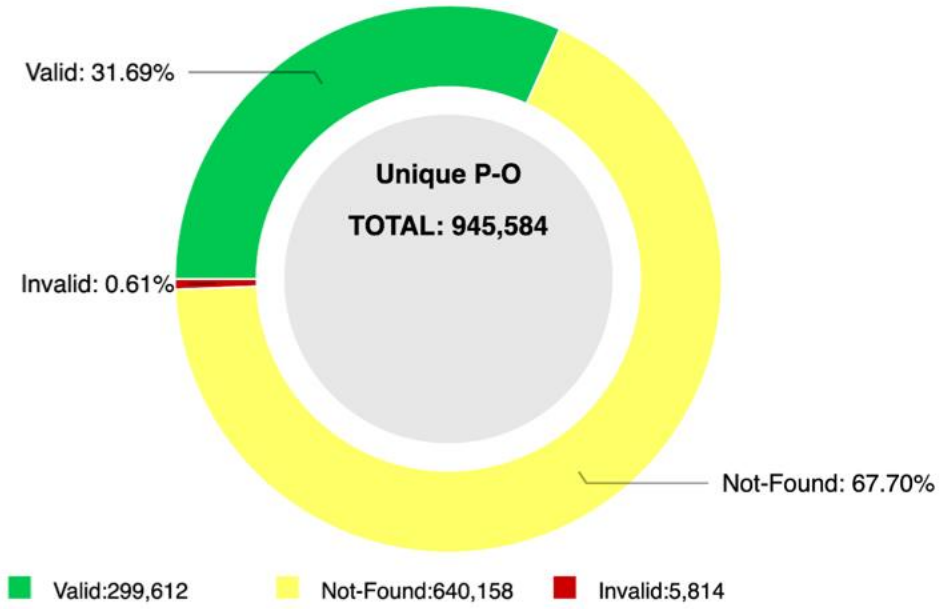
Two necessary actions

Zayo needs to use RPKI to filter BGP

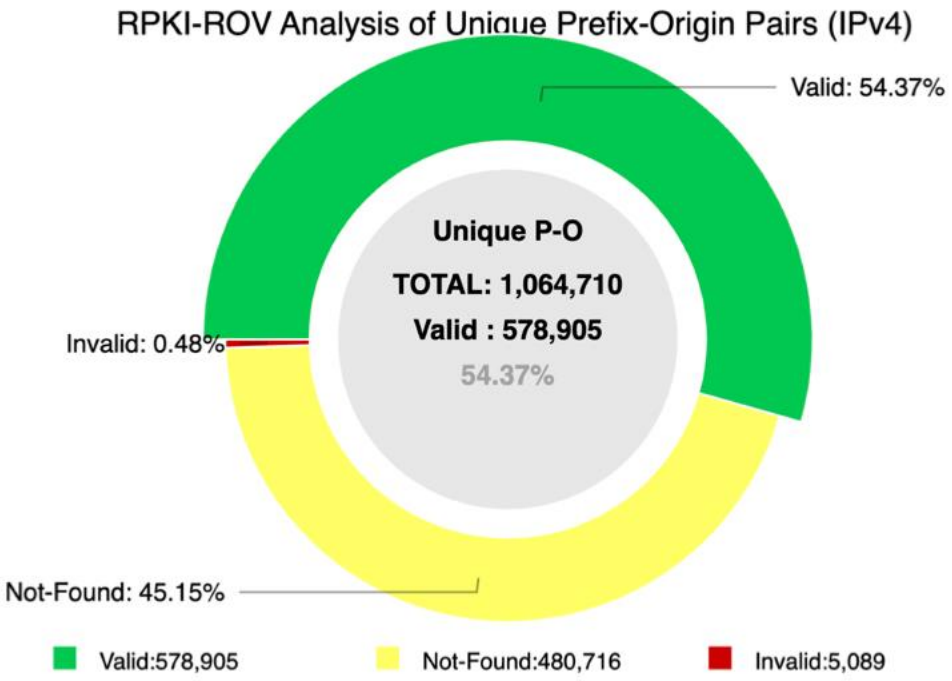


Google needs to register the RPKI record

RPKI Deployment

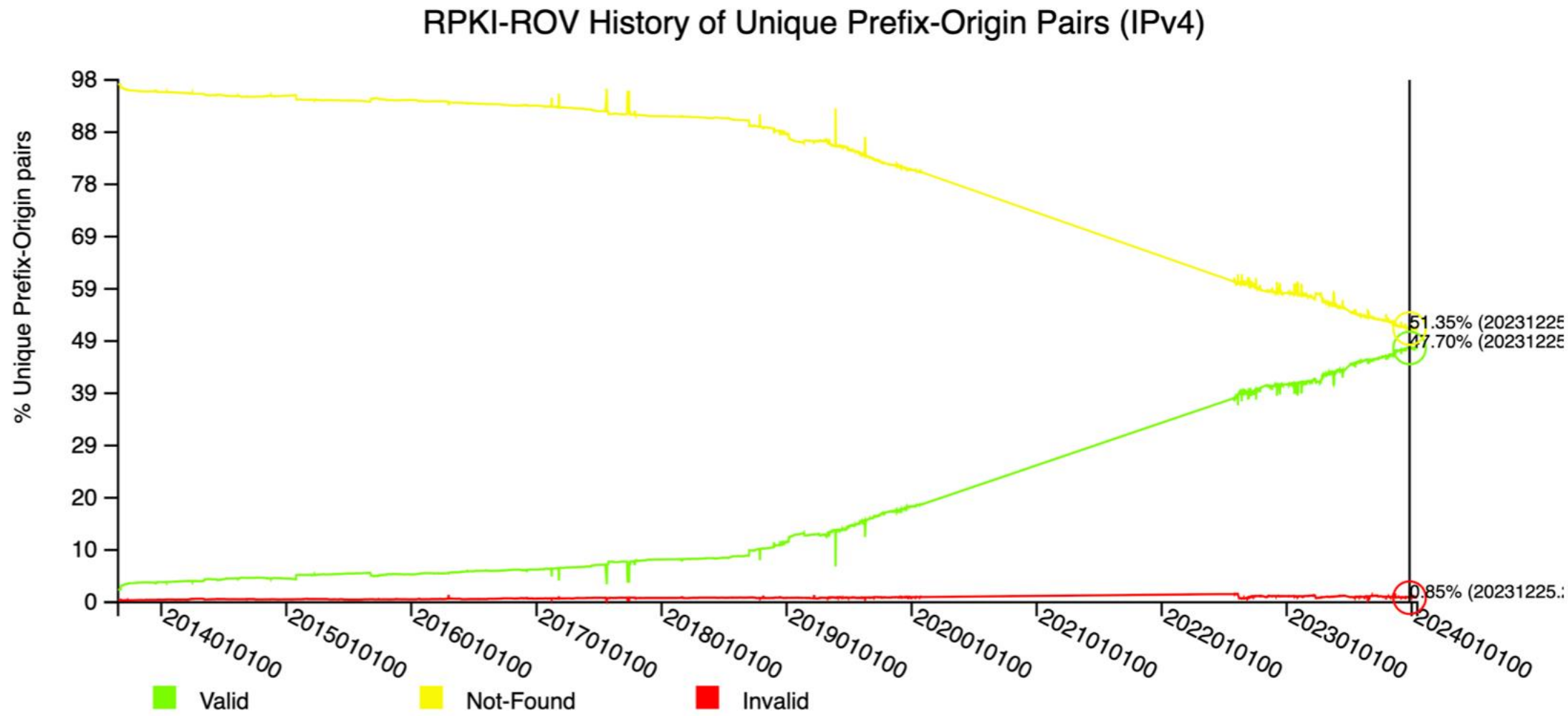


2023



Today

RPKI Deployment History



NIST RPKI Monitor: RPKI-ROV Analysis **Protocol:** IPv4 **RIR:** All